Exercise J-M

1. If A and B are two non-zero n × n matrics such that $A^2 + B = A^2 B$, then : **[JEE(Main)Jan.2023]**

यदि A तथा B, n × n के दो शून्येत्तर आव्यूह इस प्रकार हैं कि $A^2 + B = A^2 B$ है, तो : **[JEE(Main)Jan.2023]**

(A) $A^2B = BA^2$ (B) $A^2 = I$ or $B = I$ (C) $AB = I$ (D) $A^2B = I$

Ans. A

Sol. $A^2 + B = A^2 B$

$(A^2 - I)(B - I)$ .....(1)

$A^2 + B = A^2 B$

$A^2 (B - I) = B$

$A^2 B (B - I)^{-1}$

$A^2 = B (A^2 - I)$

$A^2 = BA^2 - B$

$A^2 + B = BA^2$

$A^2B = BA^2$

2. Let α be a root of the equation $(a - c)x^2 + (b - a)x + (c - b) = 0$ where a, b, c are distinct real numbers such that the matrix $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ is singular. Then, the value of $\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$ is :

[JEE(Main)Jan.2023]

माना समीकरण $(a - c)x^2 + (b - a)x + (c - b) = 0$, का मूल α है तथा आव्यूह जहाँ a, b, c भिन्न वास्तविक संख्याऐं है जबकि आव्यूह $\begin{bmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{bmatrix}$ अव्युत्क्रमणीय है। तो $\frac{(a-c)^2}{(b-a)(c-b)} + \frac{(b-a)^2}{(a-c)(c-b)} + \frac{(c-b)^2}{(a-c)(b-a)}$ का मान है :

[JEE(Main)Jan.2023]

(A) 3 (B) 12 (C) 6 (D) 9

Ans. A

Sol. $\Delta = 0 = \begin{vmatrix} \alpha^2 & \alpha & 1 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$

$\Rightarrow \alpha^2(c - b) - \alpha(c - a) + (b - a) = 0$

It is singular when $\alpha = 1$

$$\frac{(a-c)^2}{(b-a)(c-b)}+\frac{(b-a)^2}{(a-c)(c-b)}+\frac{(c-b)^2}{(a-c)(b-a)}$$

$$\frac{(a-b)^3+(b-c)^3+(c-a)^3}{(a-b)(b-c)(c-a)}$$

$$=3\frac{(a-b)(b-c)(c-a)}{(a-b)(b-c)(c-a)}=3$$

3. Let A be a 3 × 3 matrix such that $|\text{adj (adj (adj A))}| = 12^4$. Then $|A^{-1} \text{ adj } A|$ is equal to :

[JEE(Main)Jan.2023]

माना A एक 3 × 3 का आव्यूह है तथा $|\text{adj (adj (adj A))}| = 12^4$ है तब $|A^{-1} \text{ adj } A|$ बराबर है :

[JEE(Main)Jan.2023]

(A) $\sqrt{6}$ (B) 1 (C) 12 (D) $2\sqrt{3}$

Ans. D

Sol. $|\text{Adj(Adj(Adj A))}|$

$=|\text{Adj(Adj A)}|^2$

$=|\text{Adj A}|^4$

$\Rightarrow|\text{A}|^8=12^4$

$\Rightarrow|\text{A}|=12^{1/2}$

Now, $|A^{-1} \text{ Adj A}| = |A^{-1}||\text{Adj A}|$

$=\frac{1}{|\text{A}|^2}|\text{A}|^2=|\text{A}|=12^{1/2}=2\sqrt{3}$

4. The number of square matrices of order 5 with entries from the set {0, 1}, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1, is : **[JEE(Main)Jan.2023]**

कोटि 5 के वर्ग आव्यूहों, जिनके अवयव समुच्चय {0, 1} से है, प्रत्येक पंक्ति के सभी अवयवों का योग 1 है तथा प्रत्येक स्तम्भ के सभी अवयवों का योग भी 1 है, की संख्या है : **[JEE(Main)Jan.2023]**

(A) 225 (B) 125 (C) 150 (D) 120

Ans. D

Sol.

$$\begin{bmatrix} - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - \\ - & - & - & - & - & - & - & - & - & - & - \end{bmatrix}$$

In each row and each column exactly one is to be placed :

$\therefore$ No. of such matrices = $5 \times 4 \times 3 \times 2 \times 1 = 120$

Alternate :

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \rightarrow 5\text{ways} \\ \rightarrow 4\text{ways} \\ \rightarrow 3\text{ways} \\ \rightarrow 4\text{ways} \\ \rightarrow 1\text{ways} \end{matrix}$$

Step-1 : Select any 1 place for 1's in row 1. Automatically some column will get filled with 0's.

Step-2 : From next now select 1 place for 1's. Automatically some column will get filled with 0's.

$\Rightarrow$ Each time one less place will be available for putting 1's.

Repeat step-2 till last row.

Req. ways = $5 \times 4 \times 3 \times 2 \times 1 = 120$

5. Let x, y, z > 1 and $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$. Then $|\text{adj}(\text{adj}\, A^2)|$ is equal to :

[JEE(Main)Jan.2023]

माना x, y, z > 1 है तथा $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$ है। तो $|\text{adj}(\text{adj}\, A^2)|$ बराबर है :

[JEE(Main)Jan.2023]

(A) 4^8 (B) 2^8 (C) 6^4 (D) 2^4

Ans. B

Sol. $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$

$$|A| = \frac{1}{\log x \log y \log z} \begin{vmatrix} \log x & \log y & \log z \\ \log x & 2\log y & \log z \\ \log x & \log y & 3\log z \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{vmatrix}$$

$|A| = 2$

$\left|\text{adj}\left(\text{adj}\,A^2\right)\right| = |A|^8$

$= 2^8$

6. Let $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$, where $i = \sqrt{-1}$

If $M = A^TB\,A$, then the inverse of the matrix $AM^{2023}A^T$ is **[JEE(Main)Jan.2023]**

माना $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ तथा $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$ है, जहाँ $i = \sqrt{-1}$ है।

यदि $M = A^TB\,A$ है, तो आव्युह $AM^{2023}A^T$ का व्युत्क्रम है **[JEE(Main)Jan.2023]**

(A) $\begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 0 \\ 2023i & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ -2023i & 1 \end{bmatrix}$

Ans. C

Sol. $AA^T = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{-3}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$B^2 = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2i \\ 0 & 1 \end{bmatrix}$

$B^3 = \begin{bmatrix} 1 & -3i \\ 0 & 1 \end{bmatrix}$

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$B^{2023} = \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$

$M = A^TBA$

$M^2 = M \cdot M = A^TBAA^TBA = A^TB^2\,A$

$M^3 = M^2 \cdot M = A^TB^2\,A\,A^TBA = A^TB^3\,A$

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$M^{2023} = \ldots\ldots\ldots\ldots A^T B^{2023} A$

$AM^{2023} A^T = AA^T B^{2023} AA^T = B^{2023}$

$= \begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$

Inverse of $(AM^{2023} A^T)$ is $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$

7. Let A, B, C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric. Consider the statements

(S1) $A^{13}B^{26} - B^{26}A^{13}$ is symmetric

(S2) $A^{26}C^{13} - C^{13}A^{26}$ is symmetric

Then: **[JEE(Main)Jan.2023]**

(A) Both S1 and S2 are true

(B) Only S1 is true

(C) Both S1 and S2 are false

(D) Only S2 is true

माना 3×3 के आव्यूहों A, B, C में A सममित है तथा B और C विषम सममित है। कथनों

(S1) $A^{13}B^{26} - B^{26}A^{13}$ सममित है

(S2) $A^{26}C^{13} - C^{13}A^{26}$ सममित है

का विचार कीजिए। तो **[JEE(Main)Jan.2023]**

(A) S1 तथा S2 दोनों सत्य है

(B) केवल S1 सत्य है

(C) S1 तथा S2 दोनों असत्य है

(D) केवल S2 सत्य है

Ans. D

Sol. $S_1 : (A^{13} B^{26} - B^{26} A^{13})^T = (A^{13} B^{26})T - (B^{26} A^{13})^T$

$\Rightarrow (B^{26})^T (A^{13})^T - (A^{13})^T (B^{26})^T$

$\Rightarrow (BT)^{26} (A^T)^{13} - (A^T)^{13} (B^T)^{26}$

$\Rightarrow (-B)^{26} A^{13} - A^{13} (-B)26$

$\Rightarrow -(A^{13}B^{26} - B26\, A^{13})$ so skew symmetric

$S_2 : (A^{26} C^{13} - C^{13} A^{26})^T$

$= (C^T)^{13} (A^T)^{26} - (A^T)^{26} (C^T)^{13}$

$= - C^{13} A^{26} + A^{26} . C^{13}$

So, symmetric matrix

Hence S_2 is true and S_1 is false

8. Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2 = 3A + \alpha I$. If $A^4 = 21A + \beta I$, then :

[JEE(Main)Jan.2023]

माना α व β वास्तविक संख्याएं है। एक 3×3 आव्यूह A है लिए $A^2 = 3A + \alpha I$. है। यदि $A^4 = 21A + \beta I$ है, तब :

[JEE(Main)Jan.2023]

(A) $\beta = 8$ (B) $\beta = -8$ (C) $\alpha = 4$ (D) $\alpha = 1$

Ans. B

Sol. $A^2 = 3A + \alpha I$

$A^3 = 3A^2 + \alpha A$

$A^3 = 3(3A + \alpha I) + \alpha A$

$A^3 = 9A + \alpha A + 3\alpha I$

$A^4 = (9 + \alpha) A^2 + 3\alpha A$

$= (9 + \alpha) (3A + \alpha I) + 3\alpha A$

$= A (27 + 6\alpha) + \alpha (9 + \alpha)$

$\Rightarrow 27 + 6\alpha = 21 \Rightarrow \alpha = -1$

$\Rightarrow \beta = \alpha (9 + \alpha) = -8$

9. The set of all values of $t \in R$ for which the matrix

$$\begin{bmatrix} e^t & e^{-1}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-1}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-t}\sin t \end{bmatrix}$$ is invertible, is : **[JEE(Main)Jan.2023]**

$t \in R$ के सभी मानों, जिनके लिए आव्यूह

$$\begin{bmatrix} e^t & e^{-1}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-1}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-t}\sin t \end{bmatrix}$$ व्युत्क्रमणीय है, का समुच्चय है : **[JEE(Main)Jan.2023]**

(A) R (B) $\{k\pi, k \in Z\}$ (C) $\left\{(2k+1)\frac{\pi}{2}, k \in Z\right\}$ (D) $\left\{k\pi + \frac{\pi}{4}, k \in Z\right\}$

Ans. A

Sol. $$|A| = \begin{bmatrix} e^t & e^{-t}(\sin t - 2\cos t) & e^{-t}(-2\sin t - \cos t) \\ e^t & e^{-t}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{bmatrix}$$

$$= e^t.e^{-t}.e^{-t}\begin{vmatrix} 1 & \sin t - 2\cos t & -2\sin t - \cos t \\ 1 & 2\sin t + \cos t & \sin t - 2\cos t \\ 1 & \cos t & \sin t \end{vmatrix}$$

$R_1 \to R_1 - R_3, R_2 \to R_2 - R_3$

$$|A| = e^{-t}\begin{vmatrix} 0 & \sin t - 3\cos t & -3\sin t - \cos t \\ 0 & 2\sin t & -2\cos t \\ 1 & \cos t & \sin t \end{vmatrix}$$

$= e^{-t}(-2\cos t(\sin t - 3\cos t) + 2\sin t(3\sin t + \cos t))$
$= e^{-t}(6\cos^2 t + 6\sin^2 t) = 6e^{-t}$
$|A| \neq 0 \ \forall \ t \in R$

10. Let $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ and $|A - d(\text{Adj } A)| = 0$. Then : **[JEE(Main)Jan.2023]**

माना $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ तथा $|A - d(\text{Adj } A)| = 0$ है। तो : **[JEE(Main)Jan.2023]**

(A) $1 + d^2 = (m + q)^2$ (B) $(1 + d)^2 = (m + q)^2$

(C) $(1 + d)^2 = m^2 + q^2$ (D) $1 + d^2 = m^2 + q^2$

Ans. B

Sol. $A = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$

$d = mq - np$

$\text{adj}A = \begin{bmatrix} q & -n \\ -p & m \end{bmatrix}$

$|A - d\,\text{Adj}\,A| = 0$

$\begin{bmatrix} m - dq & n + dn \\ p + dp & q - dm \end{bmatrix}$

$(m - dq)(q - dm) - (d + 1)^2 np = 0$
$mq - dm^2 - dq^2 + d^2 mq = (d + 1)^2 np$
$mq(1 + d^2) - d(m^2 + q^2) = (d + 1)^2 (mq - d)$
$mq(1 + d^2) - d(m^2 + q^2) = (d^2 + 2d + 1)\, mq - (d + 1)^2 d$
$(m + q)^2 = (d + 1)^2$

11. If P is 3×3 real matrix such that $P^T = aP + (a-1)I$, where a > 1, then : **[JEE(Main)Jan.2023]**

(A) |Adj P| > 1 (B) $|\text{Adj P}| = \frac{1}{2}$

(C) |Adj P| = 1 (D) P is a singular matrix

यदि P एक 3×3 का वास्तविक आव्यूह इस प्रकार है कि $P^T = aP + (a-1)I$ है, जहाँ a > 1 है, तब : **[JEE(Main)Jan.2023]**

(A) |Adj P| > 1 (B) $|\text{Adj P}| = \frac{1}{2}$

(C) |Adj P| = 1 (D) P एक singular matrix है

Ans. C

Sol. $(P^T)^T = (aP + (a-1)I)^T$

$\Rightarrow P = aP^T + (a-1)I$

$\Rightarrow P = a(ap + (a-1)I) + (a-1)I$

$\Rightarrow P = a^2P + a(a-1)I + (a-1)I$

$\Rightarrow P = a^2P + (a^2-1)I$

$\Rightarrow (a^2-1)P + (a^2-1)I = 0$

$\Rightarrow (a^2-1)(P+I) = 0$

$\therefore a > 1$

$\Rightarrow P + I = 0$

$\Rightarrow P = -I$

adj P = adj (–I)

$\Rightarrow |\text{adj P}| = |\text{adj }(-I)| = |-I|^2$

$\Rightarrow |\text{adj P}| = 1$

12. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the diagonal elements of the matrix $(A+I)^{11}$ is equal to : **[JEE(Main)Jan.2023]**

माना $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$ है। तो आव्यूह $(A+I)^{11}$ के विकर्ण के अवयवों का योग है : **[JEE(Main)Jan.2023]**

(A) 6144 (B) 4094 (C) 2050 (D) 4097

Ans. D

Sol. $A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} = A$

Now $(A + I)^{11} = {}^{11}C_0\, A^{11} + {}^{11}C_1 + {}^{11}C_1\, A^{10} + \ldots\ldots\ldots + {}^{11}C_{11}\, I$

$= A\,({}^{11}C_0 + {}^{11}C_1 + \ldots\ldots\ldots + {}^{11}C_{10}) + I$

$= A\,(2^{11} - 1) + I$

Trace of $(A + I)^{11}$

$= 2^{11} + 4(2^{11} - 1) + 1 + (-3)\,(2^{11} - 1) + 1$

$= 2 \times 2^{11} + 1$

$= 2^{12} + 1$

13. Let A be the symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s, then $\frac{\beta s}{\alpha^2}$ is equal to _____. **[JEE(Main)Jan.2023]**

माना A एक सममित आव्यूह है जिसकी सारणिक 2 है तथा $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$ है। यदि A के विकर्ण के अवयवों का योग s है, तो $\frac{\beta s}{\alpha^2}$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 5

Sol. $A^T = A \Rightarrow$ So, $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

now $ad - b^2 = 2$ ...(1)

also $BA = \begin{bmatrix} 2 & 1 \\ 3 & 3/2 \end{bmatrix}\begin{bmatrix} a & b \\ b & d \end{bmatrix}$

$\Rightarrow 2a + b = 1$ ...(2)

$3a + \frac{3b}{2} = \alpha$ ...(3)

$3b + \frac{3d}{2} = \beta$ ...(4)

$2b + d = 2$ ...(5)

Solving equations

$a = \frac{3}{4},\ \alpha = \frac{3}{2},\ b = -\frac{1}{2}$

$d = 3,\ \beta = 3$

So $\frac{\beta \times s}{\alpha} = 5$

14. Let A be a $n \times n$ matrix such that $|A| = 2$. If the determinant of the matrix $\text{Adj}(2 \cdot \text{Adj}(2A^{-1})) \cdot$ is 2^{84}, then n is equal to _____. **[JEE(Main)Jan.2023]**

माना A एक $n \times n$ आव्यूह है तथा $|A| = 2$ है। यदि आव्यूह $\text{Adj}(2 \cdot \text{Adj}(2A^{-1}))$ का सारणिक 2^{84} है, तो n बराबर है। **[JEE(Main)Jan.2023]**

Ans. 5

Sol. $|A| = 2$

$$\text{Hence Adj}(2.\text{Adj}(2A^{-1})) = \left|2.\text{Adj}(2A^{-1})\right|^{n-1}$$

$$= \left(2^n \left|\text{Adj}(2A^{-1})\right|\right)^{n-1}$$

$$= \left(2^n \left|2A^{-1}\right|^{n-1}\right)^{n-1}$$

$$= 2^{n(n-1)}\left(\left(2^n \left|A^{-1}\right|\right)^{(n-1)}\right)^{(n-1)} \quad \therefore \left|A^{-1}\right| = \frac{1}{|A|} = \frac{1}{2}$$

$$= 2^{n(n-1)}\left(\left(2^{(n-1)}\right)^{(n-1)}\right)^{(n-1)}$$

$$= 2^{n(n-1)+(n-1)^3} = 2^{84}$$

$$\Rightarrow n = 5$$

15. Let $A = [a_{ij}], a_{ij} \in Z \cap [0,4], 1 \le i, j \le 2$. The number of matrices A such that the sum of all entries is a prime number $p \in (2,13)$ is _____. **[JEE(Main)Jan.2023]**

माना $A = [a_{ij}], a_{ij} \in Z \cap [0,4], 1 \le i, j \le 2$ है। ऐसे आव्यूहों A जिनके सभी अवयवों का योग एक अभाज्य संख्या $p \in (2,13)$ है, की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 196

Sol. As given a + b + c + d = 3 or 5 or 7 or 11

if sum = 3

$$\left(1 + x + x^2 + \ldots\ldots + x^4\right)^4 \to x^3$$

$$\left(1 - x^5\right)^4 (1 - x)^{-4} \to x^3$$

$$\therefore {}^{4+3-1}C_3 = {}^6C_3 = 20$$

If sum = 5

$$\left(1 - 4x^5\right)(1 - x)^{-4} \to x^5$$

$$\Rightarrow {}^{4+5-1}C_5 - 4x\ {}^{4.4+0-1}C_0 = {}^8C_5 - 4 = 52$$

If sum = 7

$$\left(1 - 4x^5\right)(1 - x)^{-4} \to x^7$$

$\Rightarrow {}^{4+5-1}C_4 - {}^{4.4+0-1}C_0 = {}^8C_5 - 4 = 52$

If sum = 11

$$\left(1-4x^5+6x^{10}\right)\left(1-x\right)^{-4} \rightarrow x^{11}$$

$\Rightarrow {}^{4+11-1}C_{11} - 4.\,{}^{4+6-4}C_6 + 6.\,{}^{4+1-1}C_1$

$= {}^{14}C_{11} - 4.\,{}^9C_6 + 6.4 = 364 - 336 + 24 = 52$

$\therefore$ Total matrices = 20 + 52 + 80 + 52 = 204

16. If $A = \frac{1}{2}\begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$, then : **[JEE(Main)Jan.2023]**

यदि $A = \frac{1}{2}\begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$ है, तो : **[JEE(Main)Jan.2023]**

(A) $A^{30} - A^{25} = 2I$ (B) $A^{30} + A^{25} + A = I$

(C) $A^{30} + A^{25} - A = I$ (D) $A^{30} = A^{25}$

Ans. C

Sol. $$A = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \cos 60^o & \sin 60^o \\ -\sin 60^o & \cos 60^o \end{bmatrix} \quad ...(1)$$

$\because A(\alpha).A(\beta) = A(\alpha+\beta)$

So by above

$$A^{30}\left(\frac{\pi}{3}\right) = \begin{bmatrix} \cos 30\left(\frac{\pi}{3}\right) & \sin 30\left(\frac{\pi}{3}\right) \\ -\sin 30\left(\frac{\pi}{3}\right) & \cos 30\left(\frac{\pi}{3}\right) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$A^{25}\left(\frac{\pi}{3}\right) = \begin{bmatrix} \cos 25\left(\frac{\pi}{3}\right) & \sin 25\left(\frac{\pi}{3}\right) \\ -\sin 25\left(\frac{\pi}{3}\right) & \cos 25\left(\frac{\pi}{3}\right) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & +\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = A$$

$A^{30} + A^{25} - A = I$

17. Let $A = \left[a_{ij}\right]_{2\times 2}$, where $a_{ij} \neq 0$ for all i, j and $A^2 = I$. Let a be the sum of all diagonal elements of A and b = |A|.

Then $3a^2 + 4b^2$ is equal to : **[JEE(Main)April,2023]**

माना $A = \left[a_{ij}\right]_{2\times 2}$, जहाँ सभी i, j के लिए $a_{ij} \neq 0$ एवं $A^2 = I$ है। माना A के विकर्ण के सभी अवयवों का योग a है और b = |A| है। तब $3a^2 + 4b^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 7 (B) 3 (C) 14 (D) 4

Ans. D

Sol. Let $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$

$$A^2 = \begin{bmatrix} p & q \\ r & s \end{bmatrix}\begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} p^2+qr & pq+qs \\ rp+rs & qr+s^2 \end{bmatrix}$$

$A^2 = I$

$\Rightarrow p^2 + qr = 1 \quad q(p+s) = 0$

$r(p+s) = 0 \qquad qr + s^2 = 1$

$q \neq 0 \Rightarrow p + s = 0 \Rightarrow a = 0$

$b = |A| = ps - qr = -p^2 - qr = -1 (\because s = -p)$

$\therefore 3a^2 + 4b^2 = 4$

18. Let P be a square matrix such that $P^2 = I - P$. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $P^\alpha + P^\beta = \gamma I - 29P$ and $P^\alpha - P^\beta = \delta I - 13P$, then $\alpha + \beta + \gamma - \delta$ is equal to : **[JEE(Main)April,2023]**

माना P एक वर्ग आव्यूह है जिसके लिए $P^2 = I - P$ है। तो $\alpha, \beta, \gamma, \delta \in \mathbb{N}$ के लिए यदि $P^\alpha + P^\beta = \gamma I - 29P$ तथा $P^\alpha - P^\beta = \delta I - 13P$ है, तो $\alpha + \beta + \gamma - \delta$ बराबर है : **[JEE(Main)April,2023]**

(A) 18 (B) 22 (C) 40 (D) 24

Ans. D

Sol. $P^2 = I - P$

$P^\alpha + P^\beta = \gamma I - 29P$

$P^\alpha - P^\beta = \delta I - 13P$

$P^4 = (I - P)^2 = I + P^2 - 2P$

$P^4 = I + I - P - 2P = 2I - 3P$

$P^8 = \left(P^4\right)^2 = (2I - 3P)^2 = 4I + 9P^2 - 12P$

$= 4I + 9(I - P) - 12P$

$P^8 = 13I - 21P$

$P^6 = P^4 \cdot P^2 \qquad = (2I - 3P)(I - P)$

$= 2I - 5P + 3P^2$

$= 2I - 5P + 3(I - P)$

$= 5I - 8P$

(1) + (2) (1) − (2)

$P^8 + P^6 = 18I - 29P$ $P^8 - P^6 = 8I - 13P$

From (A) $\alpha = 8, \quad \beta = 6$

$\gamma = 18$

$\delta = 8$

$\alpha + \beta + \gamma - \delta = 32 - 8 = 24$

19. Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. If $|\text{adj}(\text{adj}(\text{adj } 2A))| = (16)^n$, then n is equal to : **[JEE(Main)April,2023]**

माना $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ है। यदि $|\text{adj}(\text{adj}(\text{adj } 2A))| = (16)^n$ है, तो n बराबर है : **[JEE(Main)April,2023]**

(A) 8 (B) 9 (C) 10 (D) 12

Ans. C

Sol. $\left|\text{adj}\left(\text{adj}\left(\text{adj } 2A\right)\right)\right|$

$= \left|\text{adj}\left(\text{adj}\left(2A\right)\right)\right|^2$

$= |\text{adj } 2A|^4$

$= |2A|^8 = (8\,|A|)^8 = 16^{10}$ $\{\because \ |A| = 4\}$

$\Rightarrow n = 10$

20. Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$. If $P^TQ^{2007}\ P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $2a + b - 3c - 4d$ equal to :

[JEE(Main)April,2023]

माना $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ तथा $Q = PAP^T$ है। यदि $P^TQ^{2007}\ P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, तो $2a + b - 3c - 4d$ बराबर है :

[JEE(Main)April,2023]

(A) 2004 (B) 2007 (C) 2006 (D) 2005

Ans. D

Sol. $PP^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I = P^T P$

$P^T Q^{2007} P = P^T \left[\left(PAP^T\right)\left(PAP^T\right).....\left(PAP^T\right) \right] P$

$= P^T \left[PA^{2007} P^T \right] P$

$= A^{2007}$

$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$A^3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

$A^{2007} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$a = 1, b = 2007, c = 0, d = 1$

$2a + b - 3c - 4d = 2005$

21. If $A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}$, $A^{-1} = \alpha A + \beta I$ and $\alpha + \beta = -2$, then $4\alpha^2 + \beta^2 + \lambda^2$ is equal to :

[JEE(Main)April,2023]

यदि $A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}$, $A^{-1} = \alpha A + \beta I$ तथा $\alpha + \beta = -2$ हैं, तो $4\alpha^2 + \beta^2 + \lambda^2$ बराबर है :

[JEE(Main)April,2023]

(A) 12 (B) 14 (C) 10 (D) 19

Ans. B

Sol. $|A - xI| = 0 \Rightarrow \begin{vmatrix} 1-x & 5 \\ \lambda & 10-x \end{vmatrix} = 0 \Rightarrow x^2 - 11x + 10 - 5\lambda = 0$

$\Rightarrow (10 - 5\lambda) A^{-1} = -A + 11I$

$\therefore \alpha = \frac{-1}{10 - 5\lambda}$ and $\beta = \frac{+11}{10 - 5\lambda}$

$\alpha + \beta = -2 \Rightarrow \frac{10}{10 - 5\lambda} = -2 \Rightarrow 10 - 5\lambda = -5 \Rightarrow \lambda = 3$

$\therefore \alpha = \frac{1}{5}$ and $\beta = \frac{-11}{5}$

$\therefore 4a^2 + \beta^2 + \lambda^2 = \frac{4}{25} + \frac{121}{25} + 3^2 = 14$ Ans.

22. If A is a 3×3 matrix and $|A| = 2$, then $\left|3\,\text{adj}\left(|3A|A^2\right)\right|$ is equal to : **[JEE(Main)April,2023]**

यदि 3×3 का एक आव्यूह A है तथा $|A| = 2$ है, तो $\left|3\,\text{adj}\left(|3A|A^2\right)\right|$ बराबर है : **[JEE(Main)April,2023]**

(A) $3^{11} \cdot 6^{10}$ (B) $3^{12} \cdot 6^{10}$ (C) $3^{10} \cdot 6^{11}$ (D) $3^{12} \cdot 6^{11}$

Ans. A

Sol. $\left|3\,\text{adj}\left(|3A|A^2\right)\right| = 3^3\left|\text{adj}\left(54A^2\right)\right| = 3^3 \cdot \left|54A^2\right|^2$

$= 3^3 \times 54^6 \times |A|^4 = 3^{11} \times 6^{10}$

23. If $A = \frac{1}{5!6!7!}\begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$, then $|\text{adj}(\text{adj}(2A))|$ is equal to : **[JEE(Main)April,2023]**

यदि $A = \frac{1}{5!6!7!}\begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$ है, तो $|\text{adj}(\text{adj}(2A))|$ बराबर है : **[JEE(Main)April,2023]**

(A) 2^{16} (B) 2^8 (C) 2^{20} (D) 2^{12}

Ans. A

Sol. $|A| = \frac{1}{5!6!7!} \times 5!6!7!\begin{vmatrix} 1 & 6 & 42 \\ 1 & 7 & 56 \\ 1 & 8 & 72 \end{vmatrix}$

$R_2 \to R_2 - R_1$ & $R_3 \to R_3 - R_2$

$|A| = \begin{vmatrix} 1 & 6 & 42 \\ 0 & 1 & 14 \\ 0 & 1 & 16 \end{vmatrix} \Rightarrow |A| = 2$

$|\text{Adj}\left(\text{adj}(2A)\right)| = |2A|^{(n-1)^2} = |2A|^4 = 2^{12}|A|^4 = 2^{16}$

24. Let A be a 2×2 matrix with real entries such that $A' = \alpha A + I$, where $\alpha \in \mathbb{R} - \{-1,1\}$. If $\det\left(A^2 - A\right) = 4$, then the sum of all possible values of α is equal to : **[JEE(Main)April,2023]**

माना A वास्तविक अवयवों का एक 2×2 आव्यूह है जिसके लिए $A' = \alpha A + I$ है, $\alpha \in \mathbb{R} - \{-1,1\}$ है। यदि $\det\left(A^2 - A\right) = 4$ है,

तो α के सभी संभव मानों का योग बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{3}{2}$ (B) 2 (C) $\frac{5}{2}$ (D) 0

Ans. C

Sol. $A^T = \alpha A + I$

$A = \alpha A^T + I$

$A = \alpha(\alpha A + I) + I$

$A = \alpha^2 A + (\alpha + 1)I$

$A\left(1-\alpha^2\right) = (\alpha+1)I$

$A = \frac{I}{1-\alpha} \ldots (1)$

$|A| = \frac{1}{(1-\alpha)^2} \ldots (2)$

$\left|A^2 - A\right| = |A||A-1| \ldots (3)$

$A - I = \frac{I}{I-\alpha} - I = \frac{\alpha}{1-\alpha} I$

$|A-I| = \left(\frac{\alpha}{1-\alpha}\right)^2 \ldots (4)$

Now $\left|A^2 - A\right| = 4$

$|A||A-I| = 4$

$\Rightarrow \frac{1}{(1-\alpha)^2} \frac{\alpha^2}{\left(1-\alpha^2\right)} = 4$

$\Rightarrow \frac{\alpha}{(1-\alpha)^2} = \pm 2$

$\Rightarrow 2(1-\alpha)^2 = \pm\alpha$

$(C_1) 2(1-\alpha)^2 = \alpha$ $(C_2) 2(1-\alpha)^3 = -\alpha$

$2\alpha^2 - 5\alpha + 2 = 0 \Rightarrow \alpha_1, \alpha_2$ $2\alpha^2 - 3\alpha + 2 = 0$

$\alpha_1 + \alpha_2 = \frac{5}{2}$ $\alpha \notin R$

25. Let $S = \left\{M = \left[a_{ij}\right], a_{ij} \in \{0,1,2\}, 1 \le i, j \le 2\right\}$ be a sample space and $A = \{M \in S : M$ is invertible$\}$ be an event. Then P(A) is equal to : **[JEE(Main)April,2023]**

माना $S=\left\{M=\left[a_{ij}\right], a_{ij} \in \{0,1,2\}, 1 \le i, j \le 2\right\}$ एक प्रतिदर्श समष्टि है तथा $A=\{M \in S: M$ व्युत्क्रमणीय है$\}$ एक घटना है। तो P(A) बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{50}{81}$ (B) $\frac{47}{81}$ (C) $\frac{49}{81}$ (D) $\frac{16}{27}$

Ans. A

Sol. $M\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a,b,c,d, \in \{0,1,2\}$

$n(s) = 3^4 = 81$

we first find $p(\overline{A})$

$|m| = 0 \Rightarrow ad = bc$

$ad = bc = 0 \Rightarrow no\cdot \text{ of } (a,b,c,d) = \left(3^2 - 2^2\right)^2 = 25$

$ad = bc = 1 \Rightarrow \text{no. of } (a,b,c,d) = 1^2 = 1$

$ad = bc = 2 \Rightarrow no\cdot \text{ of } (a,b,c,d) = 2^2 = 4$

$ad = bc = 4 \Rightarrow \text{ no. of } (a,b,c,d) = 1^2 = 1$

$\therefore P(\overline{A}) = \frac{31}{81} \Rightarrow p(A) = \frac{50}{81}$

26. Let $A=\begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$. If $B=\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}A\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50} B^n$ is equal to : **[JEE(Main)April,2023]**

माना $A=\begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$ है। यदि $B=\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}A\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$ है, तो $\sum_{n=1}^{50} B^n$ के सभी अवयवों का योग है : **[JEE(Main)April,2023]**

(A) 75 (B) 100 (C) 50 (D) 125

Ans. B

Sol. $\because \overset{(M)}{\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}}\overset{(N)}{\begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\therefore \quad MN = I = NM$

$B = MAN$

$B^n = (MAN)^n = (MAN)(MAN)\ldots$

$= \left(MA^2N\right)MAN$

$= MA^nN$

$$\text{Now } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{51} \\ 0 & 0 \end{bmatrix} = I + E$$

$E^2 = 0$

$$\therefore A^n = (I+E)^n = I + nE + \underbrace{{}^nC_2E^2}_{0}$$

$= I + nE$

$$= \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix}$$

$$\text{So, } B^n = MA^nN = \begin{bmatrix} 1+\frac{n}{51} & \frac{n}{51} \\ \frac{-n}{51} & 1-\frac{n}{51} \end{bmatrix}$$

$$\sum_{n=1}^{50} B^n = \begin{bmatrix} 50+\frac{50\cdot 51}{2\cdot 51} & \frac{50\cdot 51}{2\cdot 51} \\ \frac{-50\cdot 51}{2\cdot 51} & 50-\frac{50\cdot 51}{2\cdot 51} \end{bmatrix} = \begin{bmatrix} 75 & 25 \\ -25 & 25 \end{bmatrix}$$

$\therefore$ sum = 100

27. The number of symmetric matrices of order 3, with all the entries from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}, is:

[JEE(Main)April,2023]

कोटि 3 के सममित आव्यूहों, जिनके सभी अवयव समुच्चय {0, 1, 2, 3, 4, 5, 6, 7, 8, 9} से हैं, की संख्या है :

[JEE(Main)April,2023]

(A) 6^{10} (B) 10^6 (C) 9^{10} (D) 10^9

Ans. B

Sol. $$A = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}, a,b,c,d,e,f \in \{0,1,2,.....9\}$$

Number of matrices = 10^6

28. Let $B=\begin{bmatrix}1 & 3 & \alpha\\ 1 & 2 & 3\\ \alpha & \alpha & 4\end{bmatrix}$, $\alpha>2$ be the adjoint of a matrix A and $|A|=2$. Then $\begin{bmatrix}\alpha-2\alpha & \alpha\end{bmatrix}B\begin{bmatrix}\alpha\\ -2\alpha\\ \alpha\end{bmatrix}$ is equal to :

[JEE(Main)April,2023]

माना $B=\begin{bmatrix}1 & 3 & \alpha\\ 1 & 2 & 3\\ \alpha & \alpha & 4\end{bmatrix}$, $\alpha>2$, एक आव्यूह A का सहखण्डज है तथा $|A|=2$ है। तो $\begin{bmatrix}\alpha-2\alpha & \alpha\end{bmatrix}B\begin{bmatrix}\alpha\\ -2\alpha\\ \alpha\end{bmatrix}$ बराबर है :

[JEE(Main)April,2023]

(A) –16 (B) 16 (C) 0 (D) 32

Ans. A

Sol. $|B| = |\text{adj } A| = |A|^{3-1} = |A|^2 = 4$

$1(8-3\alpha)-3(4-3\alpha)+\alpha(\alpha-2\alpha)=4$

$\Rightarrow 8-3\alpha-12+9\alpha-\alpha^2=4$

$\Rightarrow \alpha^2-6\alpha+8=0$

$\Rightarrow \alpha=2,4$

$\Rightarrow \alpha=4 \qquad (\because \alpha>2)$

Now

$$\begin{bmatrix}4 & -8 & 4\end{bmatrix}\begin{bmatrix}1 & 3 & 4\\ 1 & 2 & 3\\ 4 & 4 & 4\end{bmatrix}\begin{bmatrix}4\\ -8\\ 4\end{bmatrix}$$

$$=16\begin{bmatrix}1 & -2 & 1\end{bmatrix}\begin{bmatrix}1 & 3 & 4\\ 1 & 2 & 3\\ 4 & 4 & 4\end{bmatrix}\begin{bmatrix}1\\ -2\\ 1\end{bmatrix}$$

$$=16\begin{bmatrix}1-2+4 & 3-4+4 & 4-6+4\end{bmatrix}\begin{bmatrix}1\\ -2\\ 1\end{bmatrix}$$

$= 16[3-6+2]$

$= 16[-1] = [-16]$

29. Let for $A=\begin{bmatrix}1 & 2 & 3\\ \alpha & 3 & 1\\ 1 & 1 & 2\end{bmatrix}$, $|A|=2$. If $|2\text{ adj }(2\text{ adj }(2A))| = 32^n$, then $3n+\alpha$ is equal to :

[JEE(Main)April,2023]

माना $A=\begin{bmatrix}1 & 2 & 3\\ \alpha & 3 & 1\\ 1 & 1 & 2\end{bmatrix}$ के लिए $|A| = 2$ है। यदि $|2 \text{ adj } (2 \text{ adj } (2A))| = 32^n$ है, तो $3n + \alpha$ बराबर है :

[JEE(Main)April,2023]

(A) 10 (B) 11 (C) 12 (D) 9

Ans. B

Sol. $A=\begin{bmatrix}1 & 2 & 3\\ \alpha & 3 & 1\\ 1 & 1 & 2\end{bmatrix}$, $|A| = 2$

$\Rightarrow |A| = 1(6-1) - \alpha(4-3) + 1(2-9) = 2$

$\Rightarrow 5 - \alpha - 7 = 2 \Rightarrow \alpha = -4$

Now $|2 \text{ adj } (2\text{adj } 2A)| = 32^n$

$\Rightarrow 2^3|2\text{adj } 2A|^2 = 32^n$

$\Rightarrow 2^3 \cdot 2^6|2A|^4 = 2^{5n}$

$\Rightarrow 2^9 \cdot 2^{12}|A|^4 = 2^{5n}$

$\Rightarrow 2^{25} = 2^{5n} \Rightarrow n = 5$

$\therefore 3n + \alpha = 15 - 4 = 11$

30. Let the determinant of a square matrix A of order m be m – n, where m and n satisfy $4m + n = 22$ and $17m + 4n = 93$. If $\det(n\,\text{adj}(\text{adj}(mA))) = 3^a5^b6^c$, then $a + b + c$ is equal to : **[JEE(Main)April,2023]**

यदि m कोटि के वर्ग आव्यूह A के सारणिक का मान m – n है, जहाँ m तथा n, $4m + n = 22$ तथा $17m + 4n = 93$ को संतुष्ट करते है। यदि $\det(n\,\text{adj}(\text{adj}(mA))) = 3^a5^b6^c$ हो, तो $a + b + c$ के मान बराबर होगा : **[JEE(Main)April,2023]**

(A) 96 (B) 101 (C) 109 (D) 84

Ans. A

Sol. $|A| = m - n$

$4m + n = 22$

$17m + 4n = 93$

$m = 5, n = 2$

$|A| = 3$

$|2 \text{ adj } (\text{adj } 5A)| = 2^5 |5A|^{16}$

$= 2^5 \cdot 5^{80} |A|^{16}$

$= 2^5 \cdot 5^{80} \cdot 3^{16}$

$= 3^{11} \cdot 5^{80} \cdot 6^5$

$a + b + c = 96$

31. Let $A=\begin{bmatrix}0 & 1 & 2\\ a & 0 & 3\\ 1 & c & 0\end{bmatrix}$, where $a,c\in\mathbb{R}$. If $A^3 = A$ and the positive value of a belongs to the interval $(n-1, n]$, where $n\in\mathbb{N}$, then n is equal to _____. **[JEE(Main)April,2023]**

माना $A=\begin{bmatrix}0 & 1 & 2\\ a & 0 & 3\\ 1 & c & 0\end{bmatrix}$, $a,c\in\mathbb{R}$ है। यदि $A^3 = A$ है तथा a का घनात्मक मान अंतराल $(n-1, n]$ में है, जहाँ $n\in\mathbb{N}$ है, तो n बराबर है। **[JEE(Main)April,2023]**

Ans. 2.00

Sol. $A^2=\begin{bmatrix}0 & 1 & 2\\ a & 0 & 3\\ 1 & c & 0\end{bmatrix}\begin{bmatrix}0 & 1 & 2\\ a & 0 & 3\\ 1 & c & 0\end{bmatrix}$

$=\begin{bmatrix}a+2 & 2c & 3\\ 3 & a+3c & 2a\\ ca & 1 & 2+3c\end{bmatrix}$

$A^3=\begin{bmatrix}a+2 & 2c & 3\\ 3 & a+3c & 2a\\ ca & 1 & 2+3c\end{bmatrix}\begin{bmatrix}0 & 1 & 2\\ a & 0 & 3\\ 1 & c & 0\end{bmatrix}$

$=\begin{bmatrix}2ca+3 & a+2+3c & 2a+4+6c\\ a^2+3ca+2a & 3+2ac & 6+3a+9c\\ a+2+3c & ca+2c+3c^2 & 2ca+3\end{bmatrix}$

$2ca+3=0, a+2+3c=1$

$a+1+3\left(\frac{-3}{2a}\right)=1$

$a+1-\frac{9}{2a}=0$

$2a^2+2a-9=0$

$a\in(1, 2]$

32. Consider the matrix $f(x)=\begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}$.

Given below are two statements :

Statement I : f(–x) is the inverse of the matrix f(x).
Statement II : f(x) f(y) = f(x + y).
In the light of the above statements, choose the correct answer from the options given below :

[JEE(Main)Jan.2024]

(A) Both statement I and statement II are false.

(B) Statement I is true but statement II is false.

(C) Both statement I and statement II are true.

(D) Statement I is false but statement II is true.

आव्यूह $f(x)=\begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}$ के लिये

नीचे दो कथन दिए गए हैं :

कथन I : आव्यूह f(x) का व्युत्क्रम f(–x) है।

कथन II : f(x) f(y) = f(x + y) है।

उपर्युक्त कथनों के लिये, नीचे दिए गए विकल्पों में से **सही उत्तर** चुनिए। **[JEE(Main)Jan.2024]**

(A) कथन I तथा कथन II दोनों असत्य हैं।

(B) कथन I सत्य है परंतु कथन II असत्य है।

(C) कथन I तथा कथन II दोनों सत्य हैं।

(D) कथन I असत्य है परंतु कथन II सत्य है।

Ans. C

Sol. $f(x)=\begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}$

$$f(x)\cdot f(x)=\begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos y & -\sin y & 0\\ \sin y & \cos y & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}\cos(x+y) & -\sin(x+y) & 0\\ \sin(x+y) & \cos(x+y) & 0\\ 0 & 0 & 1\end{bmatrix}$$

$= f(x + y)$
So, statement II is true.
$f(x)\, f(–x) = 1$
So, f(– x) is inverse of f(x)

So, statement I is true.

33. Let A be a square matrix such that $AA^T = I$. Then $\frac{1}{2}A\left[\left(A+A^T\right)^2+\left(A-A^T\right)^2\right]$ is equal to :

[JEE(Main)Jan.2024]

माना एक वर्ग आव्यूह A के लिए $AA^T = I$ है। तो $\frac{1}{2}A\left[\left(A+A^T\right)^2+\left(A-A^T\right)^2\right]$ बराबर है :

[JEE(Main)Jan.2024]

(A) $A^2 + I$ (B) $A^2 + A^T$ (C) $A^3 + I$ (D) $A^3 + A^T$

Ans. D

Sol. $AA^T = I = A^TA$

On solving given expression, we get

$$\frac{1}{2}A\left[A^2+\left(A^T\right)^2+2AA^T+A^2+\left(A^T\right)^2-2AA^T\right.$$

$$=A\left[A^2+\left(A^T\right)^2\right]=A^3+A^T$$

34. Let $A=\begin{bmatrix}1 & 0 & 0\\ 0 & \alpha & \beta\\ 0 & \beta & \alpha\end{bmatrix}$ and $|2A|^3 = 2^{21}$ where $\alpha, \beta \in Z$. Then a value of α is : **[JEE(Main)Jan.2024]**

माना $A=\begin{bmatrix}1 & 0 & 0\\ 0 & \alpha & \beta\\ 0 & \beta & \alpha\end{bmatrix}$ तथा $|2A|^3 = 2^{21}$ हैं, जहाँ $\alpha, \beta \in Z$ हैं। तो α का एक मान है : **[JEE(Main)Jan.2024]**

(A) 5 (B) 17 (C) 3 (D) 9

Ans. A

Sol. $A=\begin{bmatrix}1 & 0 & 0\\ 0 & \alpha & \beta\\ 0 & \beta & \alpha\end{bmatrix}$

$$|A|=\alpha^2-\beta^2$$

$$|2A|^3=2^{21}$$

$$\Rightarrow \quad 2^9|A|^3=2^{21}$$

$\Rightarrow \quad |A|^3 = 2^{12}$

$|A| = 2^4 \qquad \Rightarrow \quad \alpha^2 - \beta^2 = 16$

$(\alpha, \beta) = (5, 3)$

$\alpha = 5$

35. Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $|P^{-1}AP - 2I|$ is equal to :

[JEE(Main)Jan.2024]

माना $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ तथा $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$ हैं। तो $|P^{-1}AP - 2I|$ के अभाज्य गुणनखण्डों का योग है :

[JEE(Main)Jan.2024]

(A) 27 (B) 26 (C) 66 (D) 23

Ans. B

Sol. $\left|P^{-1}AP - 2I\right|$

$= \left|P^{-1}AP - 2P^{-1}P\right|$

$= \left|P^{-1}(A - 2I)P\right|$

$= \left|P^{-1}\right|\left|A - 2I\right|\left|P\right|$

$= \left|A - 2I\right|$

$= \begin{vmatrix} 0 & 1 & 2 \\ 6 & 0 & 11 \\ 3 & 3 & 0 \end{vmatrix} = 69$

So, Prime factor of 69 is 3 and 23

So, sum = 26

36. Let $R = \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$ be a non-zero 3×3 matrix, where

$x\sin\theta = y\sin\left(\theta + \frac{2\pi}{3}\right) = z\sin\left(\theta + \frac{4\pi}{3}\right) \neq 0, \theta \in (0, 2\pi)$. For a square matrix M, let trace (M) denote the sum of all the diagonal entries of M. Then, among the statements : **[JEE(Main)Jan.2024]**

(I) Trace (R) = 0

(II) If trace (adj(adj(R)) = 0 , then R has exactly one non-zero entry.

(A) Both (I) and (II) are true (B) Only (II) is true

(C) Only (I) is true (D) Neither (I) nor (II) is true

माना $R=\begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$, 3×3 का एक शून्येत्तर आव्यूह है, जहाँ

$x\sin\theta = y\sin\left(\theta+\frac{2\pi}{3}\right) = z\sin\left(\theta+\frac{4\pi}{3}\right) \neq 0, \theta \in (0,2\pi)$ एक वर्ग आव्यूह M के लिए माना trace (M), M के विकर्ण के सभी अवयवों के योग को दर्शाता है। तो कथनों :

(I) Trace (R) = 0

(II) यदि trace (adj(adj(R)) = 0 है, तो R का केवल एक अवयव शून्येत्तर है **[JEE(Main)Jan.2024]**

(A) (I) तथा (II) दोनों सत्य हैं (B) केवल (II) सत्य हैं

(C) केवल (I) सत्य हैं (D) न तो (I) नहीं (II) सत्य है

Ans. D

Sol. $x\sin\theta = y\sin\left(\theta+\frac{2\pi}{3}\right) = z\sin\left(\theta+\frac{4\pi}{3}\right) \neq 0$

$\Rightarrow x, y, z \neq 0$

Also,

$\sin\theta + \sin\left(\theta+\frac{2\pi}{3}\right) + \sin\left(\theta+\frac{4\pi}{3}\right) = 0 \forall \theta \in R$

$\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=0$

$\Rightarrow xy + yz + zx = 0$

(i) Trace (R) = x + y + z

If x + y + z = 0 and xy + yz + zx = 0

$\Rightarrow x = y = z = 0$

Statement (i) is false

(ii) Adj (Adj(R)) = |R| R

Trace (Adj(Adj(R)))

$= xyz(x+y+z) \neq 0$

Statement (ii) is also false.

37. Let A be a 3 × 3 real matrix such that $A\begin{pmatrix}1\\0\\1\end{pmatrix}=2\begin{pmatrix}1\\0\\1\end{pmatrix}, A\begin{pmatrix}-1\\0\\1\end{pmatrix}=4\begin{pmatrix}-1\\0\\1\end{pmatrix}, A\begin{pmatrix}0\\1\\0\end{pmatrix}=2\begin{pmatrix}0\\1\\0\end{pmatrix}$. Then, the system $(A-3I)\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}$ has : **[JEE(Main)Jan.2024]**

(A) no solution (B) exactly two solutions

(C) infinitely many solutions (D) unique solution

माना एक 3 × 3 वास्तविक आव्यूह A के लिए $A\begin{pmatrix}1\\0\\1\end{pmatrix}=2\begin{pmatrix}1\\0\\1\end{pmatrix}, A\begin{pmatrix}-1\\0\\1\end{pmatrix}=4\begin{pmatrix}-1\\0\\1\end{pmatrix}, A\begin{pmatrix}0\\1\\0\end{pmatrix}=2\begin{pmatrix}0\\1\\0\end{pmatrix}$ हैं। तो निकाय $(A-3I)\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}$: **[JEE(Main)Jan.2024]**

(A) कोई हल नहीं है (B) के मात्र दो हल हैं

(C) के अनंत हल हैं (D) का अद्वितीय हल है

Ans. D

Sol. $A \to 3\times 3$ matrix

$$A\begin{pmatrix}1\\0\\1\end{pmatrix}=2\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$A\begin{pmatrix}-1\\0\\1\end{pmatrix}=4\begin{pmatrix}-1\\0\\1\end{pmatrix}$$

$$A\begin{pmatrix}0\\1\\0\end{pmatrix}=2\begin{pmatrix}0\\1\\0\end{pmatrix}$$

$$\text{Then } (A-3I)\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}$$

$$\text{Let } A = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}$$

Now using given condition

$$\text{(i)} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}_{3\times3} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_{3\times1} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} y_1 = 0 \\ y_2 = 2 \\ y_3 = 0 \end{matrix}$$

$$\text{(ii)} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}_{3\times3} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \\ 4 \end{pmatrix}$$

$$-x_1 + z_1 = -4$$

$$-x_2 + z_2 = 0$$

$$-x_3 + z_3 = 4$$

$$\text{(iii)} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}_{3\times3} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$$x_1 + z_1 = 2$$

$$x_2 + z_2 = 0$$

$$x_3 + z_3 = 2$$

$$-x_1 + z_1 = -4$$

$$x_1 + z_1 = 2$$

$$\Rightarrow z_1 = -1$$

$$x_1 = 3$$

So now

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

$$A-3I=\begin{bmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{bmatrix}$$

$$(A-3I)\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1\\ 2\\ 3\end{pmatrix}$$

$$\begin{pmatrix}0 & 0 & -1\\ 0 & -1 & 0\\ -1 & 0 & 0\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1\\ 2\\ 3\end{pmatrix}$$

$-z=3\Rightarrow z=-3$

$-y=2\Rightarrow y=-2 \qquad \Rightarrow$ unlique Solution

$-x=1\Rightarrow x=-1$

38. If $A=\begin{bmatrix}\sqrt{2} & 1\\ -1 & \sqrt{2}\end{bmatrix}$, $B=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$, $C=ABA^T$ and $X=A^TC^2A$, then det X is equal to :

[JEE(Main)Jan.2024]

यदि $A=\begin{bmatrix}\sqrt{2} & 1\\ -1 & \sqrt{2}\end{bmatrix}$, $B=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$, $C=ABA^T$ तथा $X=A^TC^2A$ हैं, तो det X बराबर है :

[JEE(Main)Jan.2024]

(A) 729 (B) 891 (C) 243 (D) 27

Ans. A

Sol. $A=\begin{bmatrix}\sqrt{2} & 1\\ -1 & \sqrt{2}\end{bmatrix}\Rightarrow \det(A)=3$

$B=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}\Rightarrow \det(B)=1$

Now $C=ABA^T\Rightarrow \det(C)=(\det(A))^2\times\det(B)$

$|C|=9$

Now $|X|=|A^TC^2A|$

$=|A^T|\,|C|^2\,|A|$

$=|A|^2\,|C|^2$

$=9\times 81$

$=729$

39. Let $A=\begin{bmatrix}2 & 0 & 1\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}$, $B=[B_1, B_2, B_3]$, where B_1, B_2, B_3 are column matrics, and $AB_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$, $AB_2=\begin{bmatrix}2\\3\\0\end{bmatrix}$, $AB_3=\begin{bmatrix}3\\2\\1\end{bmatrix}$. If $\alpha=|B|$ and β is the sum of all the diagonal elements of B, then $\alpha^3+\beta^3$ is equal to _____.

[JEE(Main)Jan.2024]

माना $A=\begin{bmatrix}2 & 0 & 1\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}$, $B=[B_1, B_2, B_3]$, जहाँ B_1, B_2, B_3 स्तम्भ आव्यूह हैं तथा $AB_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$, $AB_2=\begin{bmatrix}2\\3\\0\end{bmatrix}$, $AB_3=\begin{bmatrix}3\\2\\1\end{bmatrix}$ हैं। यदि $\alpha=|B|$ तथा B के विकर्ण के सभी अवयवों का योग β है, तो $\alpha^3+\beta^3$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 28

Sol. Let $B_1=\begin{bmatrix}p\\q\\r\end{bmatrix}$

$$AB_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$\Rightarrow A_1=\begin{bmatrix}2 & 0 & 1\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}\begin{bmatrix}p\\q\\r\end{bmatrix}=\begin{bmatrix}1\\0\\0\end{bmatrix}$$

$\Rightarrow$ $2p + r = 1$

$p + q = 0$ and $p + r = 0$

$p = 1, q = -1$ and $r = -1$

$$B_1=\begin{bmatrix}1\\-1\\-1\end{bmatrix}$$

Let $B_2=\begin{bmatrix}x\\y\\z\end{bmatrix}$

$$AB_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

$\Rightarrow$ $2x + z = 2$

$x + y = 3$ and $x + z = 0$

$x = 2$, $y = 1$ and $z = -2$

$$B_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

Let $B_3 = \begin{bmatrix} \lambda \\ \mu \\ \phi \end{bmatrix}$

$$AB_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda \\ \mu \\ \phi \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$\Rightarrow$ $2\lambda + \phi = 3$

$\lambda + \mu = 2$

$\lambda + \phi = 1$

$\lambda = 2$, $\mu = 0$ and $\phi = -1$

$$B_3 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \Rightarrow B = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$\alpha = |B| = 3$

$\beta = 1 + 1 - 1 = 1$

$\alpha^3 + \beta^3 = (3)^3 + 1^3 = 28$

40. Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A - xI| = 0$ be -1 and 3 , then the sum of the diagonal elements of the matrix A^2 is_____.

[JEE(Main)Jan.2024]

माना A एक 2×2 का वास्तविक आव्यूह है तथा I, 2 कोटि का तत्समक आव्यूह है। यदि समीकरण $|A - xI| = 0$ के मूल -1 तथा 3 हैं, तो आव्यूह A^2 के विकर्ण के अवयवों का योग है _____ **[JEE(Main)Jan.2024]**

Ans. 10

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$|A - xI| = 0 \Rightarrow \begin{bmatrix} a-x & b \\ c & d-x \end{bmatrix} = 0$$

$(a - x)(d - x) - bc = 0$

$x^2 - (a + d)x + ad - bc = 0$

has roots -1 and 3

$a + d = 2$ $ad - bc = -3$

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}$$

Sum of diagonal element of A^2

$= a^2 + d^2 + 2bc$

$= (a + d)^2 + 2(bc - ad)$

$= 4 + 6 = 10$

41. Let A be a 3×3 matrix det (A) = 2. If $n = \det\underbrace{\left(\text{adj}\left(\text{adj}\left(.....\left(\text{adj}A\right)\right)\right)\right)}_{2024-\text{times}}$, then the remainder when n is divided by 9 is equal to _____. **[JEE(Main)Jan.2024]**

माना A एक 3×3 का आव्यूह है तथा $\det(A) = 2$ है। यदि $n = \det\underbrace{\left(\text{adj}\left(\text{adj}\left(.....\left(\text{adj}A\right)\right)\right)\right)}_{2024-\text{times}}$ है, तो n को 9 से विभाजित करने पर शेषफल है _____ **[JEE(Main)Jan.2024]**

Ans. 7

Sol. $|A| = 2$

$$n = |\underbrace{\text{adj}(\text{adj}(.....(\text{adj}A))}_{2024-\text{ times}}|$$

$$= |A|^{(x-1)^{2024}}$$

$$\Rightarrow |A|^{2024} \Rightarrow 2^{2^{2004}}$$

Now $2^{2024} \Rightarrow 2^2(2)^{2022}$

$$\Rightarrow 2^2\left(2^3\right)^{674}$$

$$\Rightarrow 2^2(8)^{674}$$

$\Rightarrow 4(9-1)^{674}$

$\Rightarrow 4(9-1)^{674} \rightarrow$ divided by 9. Thus remainder is 4

So, $2^{2024} \rightarrow 9K + 4(K = \text{even})$

Now $\rightarrow 2^{9K+4} \Rightarrow 16.2^{9K}$

$\Rightarrow 16\left(2^3\right)^{3K}$

$\Rightarrow 16(8)^{3K}$

$\Rightarrow 16(9-1)^{3K} \rightarrow$ even

$\Rightarrow 9\alpha + 16\{\alpha \rightarrow \text{integer}\}$

$\Rightarrow 9\alpha + 9 + 7$

$\Rightarrow 9(\alpha + 1) + 7$

remainder $\rightarrow 7$

42. Let $A = I_2 - 2MM^T$, where M is a real matrix of order 2×1 such that the relation $M^T M = I_1$ holds. If λ is a real number such that the relation $AX = \lambda X$ holds for some non-zero real matrix X of order 2×1, then the sum of squares of all possible values of λ is equal to _____. **[JEE(Main)Jan.2024]**

माना 2×1 के एक वास्तविक आव्यूह M के लिए $M^T M = I_1$ है तथा $A = I_2 - 2MM^T$ है। λ एक वास्तविक संख्या है तथा कोटि 2×1 के किसी शून्येत्तर वास्तविक आव्यूह X के लिए $AX = \lambda X$ है, तो λ के सभी संभव मानों के वर्गों का योग है _____

[JEE(Main)Jan.2024]

Ans. 2

Sol. $A = I_2 - 2MM^T$

$A^2 = (I_2 - 2MM^T)(I_2 - 2MM^T)$

$= I_2 - 2MM^T - 2MM^T + 4MM^TMM^T$

$= I_2 - 4MM^T + 4MM^T$

$= I_2$

$AX = \lambda X$

$A^2X = \lambda AX$

$X = \lambda(\lambda X)$

$X = \lambda^2 X$

$X(\lambda^2 - 1) = 0$

$\lambda^2 = 1\ \lambda^2 = 1$

$\lambda = \pm 1$

Sum of square of all possible values =2

43. Let $\alpha \in (0, \infty)$ and $A = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. If $\det(\text{adj}(2A - A^T) \cdot \text{adj}(A - 2A^T)) = 2^8$, then $(\det(A))^2$ is equal to :

[JEE(Main)April,2024]

माना $\alpha \in (0, \infty)$ है तथा $A=\begin{bmatrix}1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2\end{bmatrix}$ है। यदि $\det(\text{adj}(2A-A^T)\cdot\text{adj}(A-2A^T)) = 2^8$ है, तो $(\det(A))^2$ बराबर है :

[JEE(Main)April,2024]

(A) 49 (B) 16 (C) 1 (D) 36

Ans. B

Sol. $\left|\text{adj}\left(A-2A^T\right).\text{adj}\left(2A-A^T\right)\right| = 2^8$

$Q = 2A^T - A \Rightarrow Q^T = 2A^T - A = -P$

$|\text{adj}(P)\text{adj}(Q)| \Rightarrow |PQ| = -2^4$

$\Rightarrow |P|\,(-|P|) = -2^4 \Rightarrow |P| = 4$ and $|Q| = -4$

$|A-2A^T| = 4$

$$A-2A^T=\begin{bmatrix}1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2\end{bmatrix}-2\begin{bmatrix}1 & 1 & 0 \\ 2 & 0 & 1 \\ \alpha & 1 & 2\end{bmatrix}=\begin{bmatrix}-1 & 0 & \alpha \\ -3 & 0 & -1 \\ -2\alpha & -1 & -2\end{bmatrix}$$

$\Rightarrow |A-2A^T| = 1 + 3\alpha = 4$

$\Rightarrow \alpha = 1 \Rightarrow |A| = -4$

$\Rightarrow |A|^2 = 16$

44. Let $A=\begin{bmatrix}1 & 2 \\ 0 & 1\end{bmatrix}$ and $B = I + \text{adj}(A) + (\text{adj}\,A)^2 + ... + (\text{adj}\,A)^{10}$. Then, the sum of all the elements of the matrix B is :

[JEE(Main)April,2024]

माना $A=\begin{bmatrix}1 & 2 \\ 0 & 1\end{bmatrix}$ तथा $B = I + \text{adj}(A) + (\text{adj}\,A)^2 + ... + (\text{adj}\,A)^{10}$ हो, तो आव्यूह B के सभी अवयवों का योग है :

[JEE(Main)April,2024]

(A) –110 (B) 22 (C) –124 (D) –88

Ans. D

Sol. $\text{Adj}(A)=\begin{bmatrix}1 & -2 \\ 0 & 1\end{bmatrix}$

$(\text{Adj}\,A)^2=\begin{bmatrix}1 & -4 \\ 0 & 1\end{bmatrix}$

$(\text{Adj}A)^{10}=\begin{bmatrix}1 & -20 \\ 0 & 1\end{bmatrix}$

$$B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & -2\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & -4\\ 0 & 1\end{bmatrix}+\ldots+\begin{bmatrix}1 & -20\\ 0 & 1\end{bmatrix}$$

$$B=\begin{bmatrix}11 & -110\\ 0 & 11\end{bmatrix}$$

$\Rightarrow$ sum of elements of B

$= -88$

45. Let A and B be two square matrices of order 3 such that $|A| = 3$ and $|B| = 2$. Then

$\left|A^TA\left(\text{adj}(2A)\right)^{-1}\left(\text{adj}(4B)\right)\left(\text{adj}(AB)\right)^{-1}AA^T\right|$ is equal to : **[JEE(Main)April,2024]**

माना कोटि 3 के दो वर्ग आव्यूह A तथा B के लिए $|A| = 3$ तथा $|B| = 2$ हैं। तो

$\left|A^TA\left(\text{adj}(2A)\right)^{-1}\left(\text{adj}(4B)\right)\left(\text{adj}(AB)\right)^{-1}AA^T\right|$ बराबर है : **[JEE(Main)April,2024]**

(A) 81 (B) 108 (C) 32 (D) 64

Ans. D

Sol. $|A| = 3, |B| = 2$

$|A^TA(\text{adj}(2A))^{-1}(\text{adj}(4B))(\text{adj}(AB))^{-1}AA^T|$

$= 3 \times 3 \times |(\text{adj}(2A)^{-1}| \times |\text{adj}(4B)| \times |(\text{adj}(AB))^{-1}| \times 3 \times 3$

$\downarrow \qquad \downarrow \qquad \downarrow$

$\dfrac{1}{|\text{adj}(2A)|} \qquad 2^{12} \times 2^2 \qquad \dfrac{1}{|\text{adj}(AB)|}$

$= \dfrac{1}{2^6|\text{adjA}|} \qquad = \dfrac{1}{|\text{adjB.adjA}|}$

$= \dfrac{1}{2^6.3^2} \qquad = \dfrac{1}{2^2.3^2}$

$= 3^4.\dfrac{1}{2^6.3^2}.2^{12}.2^2.\dfrac{1}{2^2.3^2} = 64$

46. Let $\alpha\beta \neq 0$ and $A=\begin{bmatrix}\beta & \alpha & 3\\ \alpha & \alpha & \beta\\ -\beta & \alpha & 2\alpha\end{bmatrix}$. If $B=\begin{bmatrix}3\alpha & -9 & 3\alpha\\ -\alpha & 7 & -2\alpha\\ -2\alpha & 5 & -2\beta\end{bmatrix}$ is the matrix of cofactors of the elements of A, then det (AB) is equal to : **[JEE(Main)April,2024]**

माना $\alpha\beta \neq 0$ तथा $A=\begin{bmatrix}\beta & \alpha & 3\\ \alpha & \alpha & \beta\\ -\beta & \alpha & 2\alpha\end{bmatrix}$ हैं। यदि आव्यूह A के अवयवों के सहखंडों का आव्यूह $B=\begin{bmatrix}3\alpha & -9 & 3\alpha\\ -\alpha & 7 & -2\alpha\\ -2\alpha & 5 & -2\beta\end{bmatrix}$ है, तो

det (AB) बराबर है : **[JEE(Main)April,2024]**

(A) 125 (B) 64 (C) 343 (D) 216

Ans. D

Sol. cofactor of 1 –1 element = $2\alpha^2 - \alpha\beta = 3\alpha$

$\Rightarrow 2\alpha - \beta = 3$

Cofactor of 2–1 element = $-(2\alpha^2 - 3\alpha) = -\alpha$

$\Rightarrow 2\alpha - 3 = 1 \Rightarrow \alpha = 2$

$\Rightarrow \beta = 1$ $\Rightarrow |A| = \begin{vmatrix}1 & 2 & 3\\ 2 & 2 & 1\\ -1 & 2 & 4\end{vmatrix} = 6$

47. If A is a square matrix of order 3 such that det(A) = 3 and $\det\left(\text{adj}\left(-4\,\text{adj}\left(-3\,\text{adj}\left(3\,\text{adj}\left((2A)^{-1}\right)\right)\right)\right)\right) = 2^m 3^n$, then m + 2n is equal to : **[JEE(Main)April,2024]**

यदि A कोटि 3 का एक वर्ग आव्यूह इस प्रकार है कि det(A) = 3 है तथा

$\det\left(\text{adj}\left(-4\,\text{adj}\left(-3\,\text{adj}\left(3\,\text{adj}\left((2A)^{-1}\right)\right)\right)\right)\right) = 2^m 3^n$, तो m + 2n बराबर है : **[JEE(Main)April,2024]**

(A) 3 (B) 2 (C) 6 (D) 4

Ans. D

Sol. $\left|\text{adj}\left(-4\text{adj}\left(-3\text{adj}\left(3\text{adj}(2A)^{-1}\right)\right)\right)\right|$

$= \left|-4\text{adj}\left(-3\text{adj}\left(3\text{adj}(2A)^{-1}\right)\right)\right|^2$

$= (-4)^6 \left|\text{adj}\left(-3\text{adj}\left(3\text{adj}(2A)^{-1}\right)\right)\right|^2$

$= 4^6.(-3)^{12} \left|\text{adj}\left(3\text{adj}(2A)^{-1}\right)\right|^4$

$= 4^6.3^{12}.3^{24} \,|\,\text{adj}(2A)^{-1}\,|^8$

$= 4^6.3^{36} \,|(2A)^{-1}|^{16}$

$= 4^6.3^{36}.2^{-48}\left(\frac{1}{|A|}\right)^{16} = \frac{4^6.3^{36}.2^{-48}}{3^{16}} = 2^{-36}.3^{20}$

$\Rightarrow m = -36, n = 20$

$m + 2n = -36 + 40 = 4$

48. Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3, then $2a + 3b$ is equal to:

[JEE(Main)April,2024]

माना $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$ यदि $A^3 = 4A^2 - A - 21I$, जहाँ I, 3×3 का तत्समक आव्यूह है, तो $2a + 3b$ बराबर होगा :

[JEE(Main)April,2024]

(A) –9 (B) –13 (C) –12 (D) –10

Ans. B

Sol. $\begin{vmatrix} 2-\lambda & a & 0 \\ 1 & 3-\lambda & 1 \\ 0 & 5 & b-\lambda \end{vmatrix} = 0$

$\Rightarrow (2 - \lambda)(3b - 3\lambda - \lambda b + \lambda^2 - 5) - ab + a\lambda = 0$

$\Rightarrow 6b - 6\lambda - 2\lambda b + 2\lambda^2 - 10 - 3\lambda b + 3\lambda^2 + \lambda^2 b - \lambda^3 + 5\lambda - ab + a\lambda = 0$

$\Rightarrow -\lambda^3 + \lambda^2(2 + 2 + b) + \lambda(-6 - 2b - 3b + 5 + a) + 6b - 10 - ab = 0$

$\Rightarrow -\lambda^3 + \lambda^2(b + 5) + \lambda(a - 5b - 1) + 6b - 10 - ab = 0$

$\Rightarrow -A^3 + 4A^2 - A - 21I = 0$

$b + 5 = 4$ $a - 5b - 1 = -1$ $2a + 3b$

$b = -1$ $a = -5$ $\Rightarrow 2(-5) + 3(-1)$

$\Rightarrow -13$

49. Let $B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1} = A^{-1}$. If $BCB^{-1} = A$ and $C^4 + \alpha C^2 + \beta I = O$, then $2\beta - \alpha$ is equal to :

[JEE(Main)April,2024]

माना $B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$ है तथा एक 2×2 आव्यूह A के लिए $AB^{-1} = A^{-1}$ है। यदि $BCB^{-1} = A$ तथा $C^4 + \alpha C^2 + \beta I = O$ हैं, तो 2β

– α बराबर है : **[JEE(Main)April,2024]**

(A) 16 (B) 2 (C) 10 (D) 8

Ans. C

Sol. $B=\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}$

$AB^{-1} = A^{-1}$

$\Rightarrow A^2 = B$

Also, $BCB^{-1} = A$

$\Rightarrow C = B^{-1}AB$

$\Rightarrow C^4 = (B^{-1}AB)(B^{-1}AB)(B^{-1}AB)(B^{-1}AB)$

$= B^{-1}A^4B$

$= B^{-1}B^2B$

$\Rightarrow C^4 = B^2$

Also, $C^2 = (B^{-1}AB)(B^{-1}AB)$

$= B^{-1}A2B$

$= B^{-1}BB$

$\Rightarrow C^2 = B$

$\Rightarrow C^4 + \alpha C^2 + \beta I = 0$

$\Rightarrow B^2 + \alpha B + \beta I = 0$

$$B^2=\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}=\begin{bmatrix}4 & 18\\ 6 & 28\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}4 & 18\\ 6 & 28\end{bmatrix}+\alpha\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}+\begin{bmatrix}\beta & 0\\ 0 & \beta\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$\Rightarrow 4 + \alpha + \beta = 0$

and $18 + 3\alpha = 0$

$\Rightarrow \alpha = -6$

$\Rightarrow \beta = 2$

$\Rightarrow 2\beta - \alpha = 10$

50. Let A be a 3×3 matrix of non-negative real elements such that $A\begin{bmatrix}1\\1\\1\end{bmatrix} = 3\begin{bmatrix}1\\1\\1\end{bmatrix}$. Then the maximum value of det(A) is _____. **[JEE(Main)April,2024]**

माना ऋणेत्तर वास्तविक अवयवों का 3×3 का एक आव्यूह A हैं तथा $A\begin{bmatrix}1\\1\\1\end{bmatrix} = 3\begin{bmatrix}1\\1\\1\end{bmatrix}$ है। तो det(A) का अधिकतम मान है _____ **[JEE(Main)April,2024]**

Ans. 27

Sol. Let $A = \begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$

Now

$$A\begin{bmatrix}1\\1\\1\end{bmatrix} = 3\begin{bmatrix}1\\1\\1\end{bmatrix}$$

$$\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}3\\3\\3\end{bmatrix}$$

$a_{11} + a_{12} + a_{13} = 3$
$a_{21} + a_{22} + a_{23} = 3$
$a_{31} + a_{32} + a_{33} = 3$

Now for maximum value of $\det(A) = a_{ij}\begin{Bmatrix}0 & i \neq j\\ 3 & i = j\end{Bmatrix}$

$\therefore |A| = 27$

51. Let A be a square matrix of order 2 such that $|A| = 2$ and the sum of its diagonal elements is –3. If the points (x, y) satisfying $A^2 + xA + yI = 0$ lie on a hyperbola, whose transverse axis is parallel to the x-axis, eccentricity is e and the length of the latus rectum is ℓ, then $e^4 + \ell^4$ is equal to _____. **[JEE(Main)April,2024]**

माना A कोटि 2 का एक वर्ग आव्यूह है, जिसके लिए $|A| = 2$ है तथा जिसके विकर्ण के अवयवों का योग –3 है। यदि $A^2 + xA + yI = 0$ को संतुष्ट करने वाले बिन्दु (x, y) एक अतिपरवलय पर हैं, जिसका अनुप्रस्थ अक्ष x-अक्ष के समांतर है, उत्केन्द्रता e है तथा नाभिलंब जीवा की लंबाई ℓ है, तो $e^4 + \ell^4$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. Bonus

Sol. Given $|A| = 2$

trace $A = -3$

and $A^2 + xA + yI = 0$

$\Rightarrow x = 3, y = 2$

so, information is incomplete to determine eccentricity of hyperbola (e) and length of latus rectum of hyperbola (l)

52. Let A be a 2×2 symmetric matrix such that $A\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}$ and the determinant of A be 1. If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2×2, then $\alpha + \beta$ equals _____. **[JEE(Main)April,2024]**

यदि 2×2 कोटि का सममित आव्यूह A इस प्रकार है कि $A\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}$ तथा A की सारणिक 1 है। यदि $A^{-1} = \alpha A + \beta I$, जहाँ I कोटि 2×2 की तत्समक आव्यूह है, तो $\alpha + \beta$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 5

Sol. Let $A = \begin{bmatrix}a & b\\b & d\end{bmatrix}$

Now $A = \begin{bmatrix}a & b\\b & d\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}$

$a + b = 3, b + d = 7$

$ad - b^2 = 1$

$(3 - b)(7 - b) - b^2 = 1$

$21 - 10b = 1$

$\Rightarrow b = 2, a = 1, d = 5$

Now, $A^{-1} = \begin{bmatrix}5 & -2\\-2 & 1\end{bmatrix}$

$= \begin{bmatrix}\alpha & 2\alpha\\2\alpha & 5\alpha\end{bmatrix} + \begin{bmatrix}\beta & 0\\0 & \beta\end{bmatrix}$

$= \alpha + \beta = 5$

53. Let $A = \begin{bmatrix}2 & -1\\1 & 1\end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n, then n is equal to _____.

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. माना $A = \begin{bmatrix}2 & -1\\1 & 1\end{bmatrix}$ है। यदि A^{13} के मुख्य विकर्ण अवयवों का योग 3^n है, तब n बराबर होगा _____

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Ans. 7

Sol. $A^2 = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 3 & 0 \end{bmatrix} = 3\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$

$A^4 = 9\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} = 9\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$

$A^8 = 81\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = 81\begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$

$A^{12} = 9 \times 81\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} = 3^6\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A^{13} = 3^6\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$

Trace = $3^6(2 + 1) = 3^7$

54. Let A be a non - singular matrix of order 3. If $\det(3\text{adj}(2\text{adj})((\det A)(A))) = 3^{-13} \cdot 2^{-10}$ and $\det(3\text{adj}(2A)) = 2^m \cdot 3^n$, then $|3m + 2n|$ is equal to _____. **[JEE(Main)April,2024]**

माना A, कोटि 3 का एक व्युत्क्रमणीय आव्यूह है। यदि $\det(3\text{adj}(2\text{adj})((\det A)(A))) = 3^{-13} \cdot 2^{-10}$ तथा $\det(3\text{adj}(2A)) = 2^m \cdot 3^n$ हैं, तो $|3m + 2n|$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 14

Sol. $|3\text{adj}(2\text{adj}(|A|A))|$

$2|3\text{adj}(2|A|\text{adj}A)|$

$2\,4|3.2.|A|\text{adj}(\text{adj}A)|$

$3^3.2^6|A|^{12}.|A|^{2^3} = 3^{-13}.2^{-10}$

$|A|^{16} = 3^{-16}\,2^{-16}$

$|A| = \frac{1}{6}$

$|3\text{adj}(2A)| = |3.2^2\text{adj}A| = 3^3.\,2^6|A|^2$

$= 3.2^4$

$M = 4\ n = 1$

$|3m + 2n| = 14$

55. Consider the matrices : $A=\begin{bmatrix}2 & -5\\ 3 & m\end{bmatrix}, B=\begin{bmatrix}20\\ m\end{bmatrix}$ and $X=\begin{bmatrix}x\\ y\end{bmatrix}$. Let the set of all m, for which the system of equations $AX = B$ has a negative solution (i.e., $x < 0$ and $y < 0$), be the interval (a, b). Then $8\int_a^b |A|dm$ is _____.

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आव्यूहों $A=\begin{bmatrix}2 & -5\\ 3 & m\end{bmatrix}, B=\begin{bmatrix}20\\ m\end{bmatrix}$, $X=\begin{bmatrix}x\\ y\end{bmatrix}$ का विचार कीजिए। माना m के सभी मान, जिनके लिए समीकरण निकाय AX = B का ऋणात्मक हल (अर्थात, $x < 0$ और $y < 0$) का समुच्चय, अंतराल (a, b) है। तो $8\int_a^b |A|dm$ बराबर है_____.

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Ans. 450

Sol. $A=\begin{pmatrix}2 & -5\\ 3 & m\end{pmatrix}, B=\begin{pmatrix}20\\ m\end{pmatrix}$

$X=\begin{pmatrix}x\\ y\end{pmatrix}$

$2x - 5y = 20$ _____(1)

$3x + my = m$ _____(2)

$\Rightarrow y=\frac{2m-60}{2m+15}$

$y<0 \Rightarrow m\in\left(\frac{-15}{2},30\right)$

$x=\frac{25m}{2m+15}$

$x<0 \Rightarrow m\in\left(\frac{-15}{2},0\right)$

$|A| = 2m + 15$

Now, $8\int_{\frac{-15}{2}}^{0}(2m+15)dm=8\left\{m^2+15m\right\}_{\frac{-15}{2}}^{0}$

$\Rightarrow 8\left\{-\left(\frac{225}{4}-\frac{225}{2}\right)\right\}$

$=8\times\frac{225}{4}=450$